

# Pre-A CL8 global Goursat-continuation certificate

Verification-first research certificate - T0 classical fixed-background scope

**Candidate:** PA-CP1-CL8-GLOBAL-GOURSAT-CONTINUATION-v0 **Internal result:**

PA-CP1-CL8-FINITE-TRIANGLE-GOURSAT-GLOBAL-EXISTENCE-STABILITY **Parents:** PA-CP1-CL8-GOURSAT-v0,

PA-CP1-CL8-CLASSICAL-BOUNDARY-T0-LATTICE-0A2-v0 **Claim context:** C6-SPACETIME-SIGNATURE **Task:** T-054

**Date:** 2026-08-03 **Package version:** 0.1.1 (revised 2026-08-04)

## 1. Verdict

The CL8 characteristic solution does not need the former one-patch smallness condition on the whole null triangle. For the declared signs

$\chi > 0, c > 0, g > 0, \lambda \geq 0, r \in \mathbb{R},$

every compatible C1 trace pair on every finite triangle  $D_\tau$  has a unique classical integral solution. The proof shifts the potential by an explicit constant, derives a nonnegative exact null-flux ledger, obtains a uniform amplitude bound by one-dimensional energy coercivity, and then tiles the triangle with finitely many translated local Volterra patches. The same amplitude bound turns the factorial-squared Volterra comparison into a global Bessel-I0 stability estimate. Section 7.1 supplies a theorem-neutral second proof of the same finite-triangle existence conclusion by a whole-triangle clipped Bielecki fixed point and a shifted-energy first-exit contradiction.

This closes the manifest-local PA-CP1-CL8-FULL-CIRCUMFERENCE-GOURSAT-EXISTENCE gate in its declared classical fixed-background scope. It covers the inserted  $L = \pi/2$  ordered PA-H1 circumference even though the old single-patch contraction factor is larger than one.

It does not select the characteristic traces, a probability measure, or a quantum state. It does not derive the Lorentzian cone, a horizon, gravity, or the physical energy reference. CP1 and Pre-A remain open.

## 2. Prior-art and novelty boundary

Global energy continuation for defocusing semilinear waves, local characteristic rectangle constructions, one-dimensional Sobolev estimates, Bielecki-weighted contractions, clipping/first-exit arguments, and factorial/Bessel bounds for Volterra equations are established mathematics. The parent certificate already cites characteristic Goursat analysis, including Gerard and Wrochna, <<https://arxiv.org/abs/1409.6691>>. No world-first or new general PDE theorem is claimed.

The repository-specific content is narrower: the exact Q3 eight-species potential, the 1/8 physical ledger, the sharp additive coercive shift, the explicit continuation constants, and the repair of the exact PA-H1 calibration gate without changing the model or hiding a state-selection assumption.

## 3. Inherited fixed-background problem

Let

$W(z) = \sum_{e=1}^8 \left( \frac{r}{2} z_e^2 + \frac{g}{4} z_e^4 \right) + \frac{\lambda}{4} \sum_{e \sim f} (z_e - z_f)^2 (z_e^2 + z_f^2),$  \tag{3.1}

where the last sum is over the twelve undirected edges of Q3. On the inserted 1+1 Lorentzian background,

$\chi \psi_{tt} - c \psi_{xx} + \nabla W(\psi) = 0, \quad s = \sqrt{c \chi}.$  \tag{3.2}

With  $u = t + x/s, v = t - x/s$ , equation (3.2) becomes

$4 \chi \psi_{uv} + \nabla W(\psi) = 0.$  \tag{3.3}

For arbitrary finite  $\tau > 0$ , set

$D_\tau = \{u \geq 0, v \geq 0, u + v \leq 2\tau\}.$  \tag{3.4}

The supplied data are

$\psi(u, 0) = A(u), \quad \psi(0, v) = B(v), \quad A(0) = B(0) = C,$  \tag{3.5}

with  $A, B \in C^1([0, 2\tau]; \mathbb{R})$ . The theorem takes (3.1)–(3.5) as inputs. In particular it does not derive the cone or the traces.

## 4. Exact coercive shift

Write

$$\|a_r\| := \max\{-r, 0\}, \text{ and } C_* = \{2a^2 \text{ over } g\}, \text{ and } \widehat{W} = W + C_*. \tag{4.1}$$

For  $r < 0$ , each onsite term satisfies the exact identity

$$\| \{r \text{ over } 2\} z^2 + \{g \text{ over } 4\} z^4 + \{a^2 \text{ over } 4g\} = \{(gz^2 - a)^2 \text{ over } 4g\} \geq 0. \tag{4.2}$$

For  $r \geq 0$  the unshifted onsite term is already nonnegative. The Q3 locking polynomial is nonnegative when  $\lambda \geq 0$ . Summing (4.2) over eight species proves

$$\|\widehat{W}(z)\| \geq 0. \tag{4.3}$$

A second completed square gives

$$\| \{r \text{ over } 2\} z^2 + \{g \text{ over } 4\} z^4 \geq \{g \text{ over } 8\} z^4 - \{a^2 \text{ over } 2g\}. \tag{4.4}$$

Consequently

$$\|\widehat{W}(z)\| \geq \{g \text{ over } 8\} \sum_e z_e^4 - \{2a^2 \text{ over } g\}. \tag{4.5}$$

The shift changes no equation because  $\text{grad } \widehat{W} = \text{grad } W$ . It is a proof normalization, not an observed vacuum energy or a comparison with physical empty space.

## 5. Nonnegative null-flux ledger

For  $0 < t \leq \tau$ , define

$$\|\widehat{E}(t) = \{1 \text{ over } 8\} \int_{-st}^{st} \left( \|\chi\| \text{ over } 2 \right) \|\psi_t\|^2 + \{c \text{ over } 2\} \|\psi_x\|^2 + \widehat{W}(\psi) \right) dx. \tag{5.1}$$

The parent current identity is unchanged by a constant potential shift. Its exact triangle integral is

$$\|\widehat{E}(t) = \{s \text{ over } 8\} \int_0^{2t} \left( \|\chi|A|\|^2 + \{\widehat{W}(A) \text{ over } 2\} \right) du + \{s \text{ over } 8\} \int_0^{2t} \left( \|\chi|B|\|^2 + \{\widehat{W}(B) \text{ over } 2\} \right) dv. \tag{5.2}$$

The shift is consistent on both sides: the slice receives  $(1/8)(2st)C_*$ , while the two null sides together receive  $2(s/8)(2t)(C_*/2)$ , the same number.

Set the total available boundary flux

$$\|\mathcal{F}_\tau = \{s \text{ over } 8\} \int_0^{2\tau} \left( \|\chi|A|\|^2 + \{\widehat{W}(A) \text{ over } 2\} \right) du + \{s \text{ over } 8\} \int_0^{2\tau} \left( \|\chi|B|\|^2 + \{\widehat{W}(B) \text{ over } 2\} \right) dv. \tag{5.3}$$

Equations (4.3) and (5.2) give the uniform finite-time bound

$$\|0 \leq \widehat{E}(t) \leq \mathcal{F}_\tau. \tag{5.4}$$

## 6. Explicit amplitude bound

Let  $I_t = [-st, st]$ . The spatial-gradient part of (5.1) gives

$$\|\|\psi_x(t)\|_{L^2(I_t)}^2 \leq \{16 \mathcal{F}_\tau \text{ over } c\}. \tag{6.1}$$

The right endpoint of the slice lies on the supplied null trace:

$$\|\psi(t, st) = A(2t). \tag{6.2}$$

For every component and every  $x$  in  $I_t$ , the fundamental theorem of calculus and Cauchy--Schwarz therefore give

$$\|\|\psi_e(t, x)\| \leq \|A\|_{L^\infty(\ell^\infty_8)} + \sqrt{2st} \|\psi_x(t)\|_2. \tag{6.3}$$

Using either null endpoint symmetrically, set

$$\|M_\partial = \max\{\|A\|_\infty, \|B\|_\infty\}. \tag{6.4}$$

Equations (6.1)–(6.4) imply the uniform max-ball

$$\|\sup_{D_\tau} \|\psi\|_{\ell^\infty_8} \leq S_\tau = M_\partial + 4\sqrt{2s\tau \mathcal{F}_\tau \text{ over } c}. \tag{6.5}$$

The quartic inequality (4.5) additionally controls the slice  $L^4$  norm, but the endpoint-anchored estimate (6.5) is already sufficient for continuation. It also avoids any degenerating interval constant near  $t=0$ . This is the a priori estimate that the one-patch argument lacked.

## 7. Uniform shell continuation

Let  $D_T$  be a maximal locally constructed triangle, with  $T < \tau$ . For a point in the new shell  $D_{(T+\delta)} \setminus D_T$ , its Volterra rectangle has an unknown cap of area

$$\|\text{area}(\text{cap}) \leq \{(u+v-2T)^2 \text{ over } 2\} \leq 2\delta^2. \tag{7.1}$$

Let

$$\|M_0\|_{A+B-C} \leq \infty, \quad K_0 = M_0 + \frac{\tau^2 S}{4\chi}, \quad R_c = K_0 + \rho, \quad \text{tag}\{7.2\}$$

where  $\rho > 0$  and the parent functions are

$$b_R = |r|R + (g + 12\lambda)R^3, \quad \ell_R = |r| + (3g + 36\lambda)R^2. \quad \text{tag}\{7.3\}$$

The part of the integral inside  $D_T$  is known and is bounded by  $K_0$ . The unknown cap defines a self-map and contraction whenever

$$\frac{\delta^2 b_R}{2\chi} \leq \rho, \quad \frac{\delta^2 \ell_R}{2\chi} < 1. \quad \text{tag}\{7.4\}$$

The constants in (7.4) do not depend on  $T$ , so a positive common  $\delta$  exists. Moreover, (6.5) bounds the field at the frontier, and the integral derivative identities

$$\|\psi_u = A'(u) - \frac{1}{4\chi} \int_0^u \nabla W(\psi(u, y)) dy, \quad \psi_v = B'(v) - \frac{1}{4\chi} \int_0^u \nabla W(\psi(x, v)) dx \quad \text{tag}\{7.5\}$$

give uniform first-derivative bounds. The solution extends continuously to the frontier and (7.4) adds another shell, contradicting maximality. Thus it exists on all of  $D_\tau$ . A causally ordered translated-rectangle tiling is an equivalent finite construction; uniqueness makes overlaps and admissible tiling orders agree.

The parent variational symplectic current extends across the finite solution because its linearized coefficients stay bounded on the same max-ball. When the domain is decomposed into shells or translated rectangles, every internal interface occurs twice with opposite orientation. Those fluxes cancel, leaving exactly the two supplied null sides and the final-slice identity. This is an inherited classical flux statement, not a state-selection theorem.

The parent factorial-squared argument makes any two continuous solutions on their compact common range identical.  $C_1$  traces give continuous first and mixed derivatives through (7.5) and the equation. Higher trace regularity bootstraps as in the parent certificate.

## 7.1 Alternate whole-triangle clipped-Bielecki audit

This lemma is an alternate proof of the finite-triangle existence and uniqueness conclusion already recorded in EXP-000734. The alternate route is recorded in EXP-000737. It introduces no new theorem, result ID, gate closure, data class, or physical scope.

Fix any  $R_{\text{bar}} > S_\tau$  and let  $\kappa_{\text{bar}} : R_8 \rightarrow [-R_{\text{bar}}, R_{\text{bar}}]^8$  be componentwise clipping. Define

$$F_{\text{bar}}(z) = \nabla W(\kappa_{\text{bar}}(z)), \quad G(u, v) = A(u) + B(v) - C. \quad \text{tag}\{7.6\}$$

Componentwise clipping is one-Lipschitz in the max norm. The registered Hessian row bound on the cube therefore gives

$$\|F_{\text{bar}}(z) - F_{\text{bar}}(\tilde{z})\| \leq \ell_{\text{bar}} \|z - \tilde{z}\|, \quad \ell_{\text{bar}} = |r| + (3g + 36\lambda)R_{\text{bar}}^2. \quad \text{tag}\{7.6\}$$

Moreover,  $F_{\text{bar}}$  is globally bounded. Hence the clipped Volterra map

$$(\mathcal{T}_{\text{bar}} \phi)(u, v) = G(u, v) - \frac{1}{4\chi} \int_0^u \int_0^v F_{\text{bar}}(\phi(\sigma, \nu)) d\nu d\sigma \quad \text{tag}\{7.7\}$$

acts on the whole Banach space  $C(D_\tau; R_8)$ , without a whole-triangle sup-ball self-map condition.

For  $\beta > 0$ , equip this space with the equivalent Bielecki norm

$$\|\phi\|_\beta = \sup_{D_\tau} e^{-\beta(u+v)} \|\phi(u, v)\|. \quad \text{tag}\{7.8\}$$

The exact weighted kernel is

$$e^{-\beta(u+v)} \int_0^u \int_0^v e^{\beta(\sigma+\nu)} d\nu d\sigma = \frac{(1 - e^{-\beta u})(1 - e^{-\beta v})}{\beta^2} \leq \frac{1}{\beta^2}. \quad \text{tag}\{7.8\}$$

Consequently,

$$\|\mathcal{T}_{\text{bar}} \phi - \mathcal{T}_{\text{bar}} \tilde{\phi}\|_\beta \leq \frac{\ell_{\text{bar}}}{4\chi} \beta^{-2} \|\phi - \tilde{\phi}\|_\beta. \quad \text{tag}\{7.9\}$$

Since  $g > 0$  and  $R_{\text{bar}} > S_\tau \geq 0$ , one has  $\ell_{\text{bar}} > 0$ . Choosing  $\beta = \sqrt{\ell_{\text{bar}} / (2\chi)}$  gives contraction factor  $1/2$ , so Banach's theorem supplies a unique clipped solution on all of  $D_\tau$ .

It remains to show that clipping never activates. If activation occurred, let  $t_* \leq \tau$  be the first slice time for which

$$\sup_{D_{t_*}} \|\psi_{\text{bar}}\| = R_{\text{bar}}.$$

Because  $M_{\text{partial}} \leq S_\tau < R_{\text{bar}}$ , one has  $t_* > 0$ . On all of  $D_{(t_*)}$ , including the first-contact set,  $\kappa_{\text{bar}} \psi_{\text{bar}} = \psi_{\text{bar}}$ ; thus the clipped solution satisfies the original equation there. The shifted-energy argument of Sections 4--6 applies and yields

$$\|\sup_{D_{\{t_*\}}}\|\psi_{\{\bar{R}\}}\|_{\infty}\leq S_{\tau}<\bar{R},\]$$

contradicting first contact. Hence clipping is everywhere inactive, and the clipped fixed point is the original solution. Conversely, every original solution obeys the same amplitude bound and is therefore a fixed point of (7.7), so clipped uniqueness also gives original uniqueness.

This alternate audit stops at the C1 existence level. Every  $C^8 \times C^8 \rightarrow C^7 \times C^6$  assertion continues to rely exclusively on the corrected Section 8.1 recursion

$$\|D_m\|\leq\max\{D_{m-1},T_m+\tau P_{m-1}/(2\chi),P_{m-2}/(4\chi)\},\]$$

with its final entry omitted at  $m=1$ , as required by EXP-000735. The contraction factor can be fixed at  $1/2$ , but  $R_{\text{bar}}$ ,  $\text{ell}_{\text{Rbar}}$ ,  $\beta$ , and the equivalence constant between the Bielecki and sup norms remain finite-horizon and trace dependent.

## 8. Global field-value stability

Let  $\psi$  and  $\tilde{\psi}$  be solutions for two compatible trace pairs. Put

$$\|R\|=\max\{S_{\tau}(A,B),S_{\tau}(\widetilde{A},\widetilde{B})\},\quad \text{ell}_R=|r|+(3g+36\lambda)R^2.\tag{8.1}\]$$

For the two free lifts, let

$$\|\delta_G\|=\|A+B-C-(\widetilde{A}+\widetilde{B}-\widetilde{C})\|_{\infty}.\tag{8.2}\]$$

Iterating the Volterra difference inequality yields

$$\|\psi-\widetilde{\psi}\|(u,v)\leq\delta_G\sum_{n=0}^{\infty}\frac{\bigl(\text{ell}_{Ruv}/(4\chi)\bigr)^n}{n!}.\tag{8.3}\]$$

Because  $uv\leq\tau^2$  on  $D_{\tau}$ , the exact Bessel majorant is

$$\|\psi-\widetilde{\psi}\|_{\infty}\leq\delta_G\,I_0\!\left(\tau\sqrt{\text{ell}_R/\chi}\right).\tag{8.4}\]$$

No whole-triangle smallness condition occurs in (8.4). This certificate claims field-value stability. Derivative and phase-space stability need higher trace norms and are not silently included.

### 8.1 High-regularity phase-map lemma

For the composition route, now assume  $A,B\in C^8$ . Let

$$\|D_m\|=\max_{p+q\leq m}\|\partial_u^p\partial_v^q\psi\|_{\infty},\quad T_m=\max\{\|A\|_{C^m},\|B\|_{C^m}\}.\tag{8.5}\]$$

Let  $P_j(D_0,\dots,D_j)$  denote the Faà di Bruno upper polynomial for  $j$  derivatives of  $\text{grad } w(\psi)$  on the  $S_{\tau}$  ball. Because  $\text{grad } w$  is cubic, its derivative tensors of order four and higher vanish, and every monomial in  $P_j$  contains at most three field derivatives whose total order is  $j$ . Differentiating the two pure Volterra formulas and the mixed equation gives

$$\|D_m\|\leq\max\left\{D_{m-1},T_m+\frac{\tau}{2\chi}P_{m-1},\frac{1}{4\chi}P_{m-2}\right\},\tag{8.6}\]$$

with the third entry omitted for  $m=1$ . The explicit  $D_{(m-1)}$  entry retains all lower derivatives already included in the definition of  $D_m$ . Thus the right side for  $D_m$  depends only on  $D_0,\dots,D_{(m-1)}$ , so (8.6) closes inductively through  $m=8$ .

For two trace pairs, subtract the same recursions. Polynomial Lipschitz bounds on a common derivative ball, starting from (8.4), give continuity at each successive order. Hence

$$\|(A,B)\mapsto(q,\Pi):C^8\times C^8\mapsto C^7\times C^6\tag{8.7}\]$$

is continuous. Every compact  $C^8$  source subset therefore has common trace, flux, amplitude, and derivative bounds. On its seam-compatible trace image, these are the uniform fixed-family inputs required by the inherited  $O(a^2)$  composition theorem. This is a high-regularity classical statement, not derivative stability for merely  $C^1$  traces.

## 9. Full ordered PA-H1 circumference

The inserted tangent calibration fixes

$$\|L=\{\pi\over 2\},\quad \{c\over\chi\}=1,\quad \{-2r\over\chi\}=9.\tag{9.1}\]$$

Thus  $s=1$  and the direct characteristic slice has  $\tau=L/(2s)=\pi/4$ . The former whole-triangle factors remain

$$\|q\geq 9\pi^2\over 32>1,\quad q_{\text{rm shifted}}\geq 9\pi^2\over 64>1.\tag{9.2}\]$$

Equation (9.2) is not repaired by pretending it is small. It is bypassed by the energy continuation theorem, which applies to every finite  $\tau$  and every compatible  $C^1$  trace pair with finite  $F_{\tau}$ . It therefore includes nonconstant smooth perturbations of the ordered equilibrium on the full inserted circumference.

As an illustrative constant-curvature **linearized control only**, splitting  $\tau$  into two equal shells makes the cap factor for  $\text{ell}=9\chi$  equal to  $9\pi^2/128<1$ , while three equal shells make the  $\text{ell}=18\chi$  factor  $\pi^2/16<1$ . These are not nonlinear

or universal patch counts. The actual nonlinear continuation uses the trace-dependent  $b_-(R_c)$ ,  $ell_-(R_c)$ , and self-map reserve in (7.2)–(7.4).

For example, on the collective branch let

$$\{q_e(x) = v + \epsilon \cos(4x), \quad \text{where } \pi_e(x) = 0, \quad 0 < |\epsilon| < \sqrt{-r/g}, \quad \text{tag}\{9.3\} \}$$

for all eight species on the  $L = \pi/2$  circle. The inherited global periodic classical Cauchy theorem evolves these smooth data backward from  $t = \tau$ . Restricting that solution to the two null sides produces compatible, nonconstant traces, and Goursat uniqueness reconstructs the same periodic final phase. More generally, take the trace image of an open periodic  $C^8 \times C^7$  Cauchy neighborhood and equip that image only with the pullback/relative topology from its Cauchy data. No independent embedded-manifold or openness theorem inside unconstrained trace space is claimed. On each compact source subset, differentiated Volterra induction gives common  $C^8$ -to- $C^7 \times C^6$  phase-map bounds as well as common flux and amplitude bounds.

Arbitrary Goursat traces do not automatically lie in that trace image. Literal direct-circle composition still requires

$$\{ \partial_x^m q(-s\tau) = \partial_x^m q(s\tau), \quad 0 \leq m \leq 7, \quad \partial_x^m \pi(-s\tau) = \partial_x^m \pi(s\tau), \quad 0 \leq m \leq 6. \quad \text{tag}\{9.4\} \}$$

At first order this includes  $A(2\tau) = B(2\tau)$  and the integral-corrected derivative conditions already derived in the composition certificate. The registered NG-2026-08-03-PRE-A-CP1-CL8-UNMATCHED-PERIODIC-COMPOSITION remains in force for generic admitted traces; global existence does not weaken that no-go.

This coverage does not derive the PA-H1 circumference, the tuning, the collective species, the characteristic role, or a state on the trace space.

## 10. Gate resolution and remaining route

The child resolution is:

- PA-CP1-CL8-FULL-CIRCUMFERENCE-GOURSAT-EXISTENCE: **closed in the declared classical fixed-background scope;**
- PA-CP1-CL8-PREFERRED-STATE-COMPOSITION-SELECTION: **open.**

The first statement means finite-triangle existence, uniqueness, explicit amplitude control, and field-value stability, including the inserted PA-H1 length. It does not mean a finite- $a$  Goursat scheme, a full-dimensional Einstein characteristic theorem, or a physical state.

EXP-000736 has since split the second gate: finite-regulator classical invariant measures exist, but invariance, the declared symmetries, smooth seam support, and regulator compatibility do not uniquely select a preferred one. The parent preferred-state gate therefore remains open. Its next bounded child is PA-CP1-CL8-FINITE-REGULATOR-QUANTUM-STATE, which must declare the exact CCR normalization and  $\hbar$  before any quantum state is imported. It must not use the arbitrary proof shift  $C_*$  as an absolute vacuum-energy choice.

## 11. Devil's-advocate review

1. **\*\*Adding  $C_*$  changes the dynamics. DISMISSED.\*\*** Its gradient is zero. The slice and both null ledgers acquire exactly the same geometric constant.
2. **The shifted energy proves energy below physical empty space. UPHELD AS FALSE.** The shift is an arbitrary proof normalization and no physical reference state is selected.
3. **A negative quadratic term can cause finite-time blow-up. DISMISSED FOR THE DECLARED SIGN CLASS.** Positive  $g$  and nonnegative  $Q_3$  locking make  $W$  bounded below and give (4.5); (6.5) prevents amplitude blow-up on every finite triangle.
4. **The local patch radius grows after every tile. DISMISSED.** The exact flux identity restores the same global bound  $S_\tau$  after each causal band, and the shell radius  $R_c$  is computed once from that bound.
5. **Energy control alone does not provide a continuation trace. DISMISSED.** The Volterra derivative formulas (7.5), the amplitude bound, and bounded boundary derivatives give uniform Lipschitz control up to a finite frontier.
6. **The theorem also works for  $g \leq 0$  or  $\lambda < 0$ . UPHELD AS FALSE.** Those signs remove the declared coercive argument and are excluded.
7. **Finite  $\tau$  implies a uniform cosmological history. UPHELD AS FALSE.**  $S_\tau$  and the stability factor may grow with  $\tau$ ; no infinite-time or thermodynamic bound is claimed.
8. **Field-value stability is full phase-space stability. UPHELD AS FALSE.** Derivative stability requires stronger trace norms.
9. **Full PA-H1 length closes the state bridge. UPHELD AS FALSE.** The trace pair and every measure or quantum state remain inputs or open gates.
10. **Existence makes arbitrary final slices periodic. UPHELD AS FALSE.** Only the seam-compatible trace subset composes directly; generic traces still require the recorded extension branch.

11. **This proves C6, CP1, or Pre-A. UPHOLD AS FALSE.** The background cone, physical state/reference, quantum composition, phase transition, cooling, gravity, and emergence links remain open.
12. **Componentwise clipping destroys the gradient structure, so the shifted energy identity cannot be applied to the clipped equation. VALID WITH MITIGATION.** In general  $F_{\text{Rbar}} = \text{grad } W$  composed with  $\kappa_{\text{Rbar}}$  need not be the gradient of a scalar potential, and no energy identity is asserted after clipping activates. The energy estimate is used only on the closed pre-exit triangle, where  $\kappa_{\text{Rbar}} \psi = \psi$  and  $F_{\text{Rbar}}(\psi) = \text{grad } W(\psi)$ , including at first contact. The strict bound  $S_{\text{tau}} < R_{\text{bar}}$  rules out that contact, so the non-gradient region is never entered.

## 12. Reproduction

Run from the repository root:

```
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/pre_a_cp1_cl8_global_goursat_continuation.py
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/pre_a_cp1_cl8_global_goursat_continuation_independent.py
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/pre_a_cp1_cl8_global_goursat_continuation_verify.py
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/pre_a_cp1_cl8_global_goursat_continuation_verify.py --check-stored
E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/pre_a_cp1_cl8_global_goursat_continuation_pdf.py
```

The primary implementation uses symbolic polynomial identities. The non-importing implementation reconstructs the Q3 counts, rational completed squares, continuation fixture, and Volterra coefficients without importing the primary module. The integrated verifier reruns both into temporary files, compares fresh and stored artifacts, audits parent hashes, scope flags, certificate anchors, and independent-source separation.

## 13. No-overclaim boundary

This certificate proves only global classical existence, uniqueness and field-value stability on each finite null triangle of an inserted fixed 1+1 CL8 theory and exhibits a nonconstant seam-compatible periodic trace family. It does not make arbitrary Goursat traces periodic, derive the cone, a horizon, spacetime, gravity, full 3+1 dynamics, a finite-regulator characteristic scheme, a preferred boundary measure, a quantum or Hadamard state, physical empty space, an absolute energy zero, a below-empty-space sign, cooling, a phase transition, C6 advancement, CP1, or Pre-A.

The clipped-Bielecki lemma is only an alternate verification of the finite-triangle classical statement already recorded in EXP-000734. It does not alter EXP-000735, provide a horizon-uniform bound as  $\tau$  tends to infinity, make generic final traces periodic, select a seam, extension, classical measure, quantum or Hadamard state, or physical vacuum, derive the inserted Lorentzian background or PA-H1 parameters, reach full 3+1 dynamics, advance C6, complete CP1 or Pre-A, or create a new theorem, result, or gate.